Finite transition-probability witnesses separating real and complex qutrits

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We construct two finite, dimension-bounded witnesses that separate complex from real qutrit geometry in a calibrated prepare-and-projective-test scenario. Twenty and twenty-one labeled rays are used both as pure-state preparations and as rank-one tests, so their transition probabilities are squared ray overlaps. The zero patterns form faithful orthogonality hypergraphs with, respectively, 12 and 14 complete qutrit contexts. Complex qutrits reproduce both tables exactly; no calibrated reciprocal real-qutrit model made of pure states and rank-one tests does. The 20-ray obstruction is the real rigidity of two rank-one decompositions of one positive operator. The 21-ray obstruction survives when the relevant pairs are orthogonal and instead follows from cyclic closure; its off-diagonal transition probabilities take only the values 0, 1/4, and 1/2. Compactness gives a strictly positive uniform gap from this real model class, turning exact nonexistence into finite noise-robust field witnesses. The result is qutrit-dimension-bounded: after dimension doubling, phase-adjusted realification preserves only the underlying orthogonality pattern in R^6; it need not reproduce nonzero transition probabilities, and the qutrit contexts cease to be maximal.

I. INTRODUCTION

A qutrit ray can be used in two reciprocal laboratory roles: as a prepared pure state and as the accepting outcome of a rank-one projective test. If the rays labeled u and v are represented by projectors P_u and P_v, the transition probability is

$$p(v|u) = \mathrm{Tr}(P_u P_v). \tag{1}$$

This raises a finite operational question: can a table of such transition probabilities be reproduced by three-dimensional quantum models over both $\mathbb{R}$ and $\mathbb{C}$? We give two exact tables that complex qutrits reproduce but real qutrits cannot.

The dimension restriction is essential. If a change of dimension and an additional complex structure are allowed, a complex Hilbert space $\mathbb{C}^n$ can be represented on $\mathbb{R}^{2n}$ by separating real and imaginary parts; related constructions reproduce broad classes of complex quantum protocols in a real formalism [1–3]. If instead one fixes the dimension, composition rules, locality assumptions, or the interpretation of a context as a maximal orthogonal family, real and complex models can be separated operationally [4–6]. Our witnesses address this latter, dimension-bounded comparison and do not purport to rule out real simulations with additional degrees of freedom.

The zero entries of a transition table form an orthogonality hypergraph: each hyperedge specifies three rays that make up a complete qutrit measurement. Harding and Salinas Schmeis gave the first finite hypergraph with a faithful orthogonal representation over $\mathbb{C}^3$ but none over $\mathbb{R}^3$ [7]. Independently, Trandafir and Cabello encountered the same field separation in a Kochen–Specker (KS) construction used to design bipartite perfect quantum strategies [8]. A subsequent analysis by Navara and Svozil relates those constructions through mutually unbiased bases and KS closures [9].

Recent extremal results also show how finite incidence structures become quantum-information resources. Cabello proved that every bipartite perfect quantum strategy defines a KS set and obtained record-small strategies when size is counted by measurement inputs [10]. A subsequent 14-basis qutrit KS set improved that record and refuted a proposed optimum [11]. Our hypergraphs are not KS sets, so that conversion is not available here. Instead, we use the complete transition table itself as the operational object.

In prepare-and-measure scenarios, a finite transition-probability witness is a table of probabilities that complex qutrits can reproduce but real qutrits cannot under the same experimental assumptions. Here we give two small exact witnesses with an octagonal incidence skeleton and, more importantly, expose two different mechanisms by which real qutrit geometry fails. The 20-vertex, 12-context hypergraph H_{20} forces an equality between two sums of rank-one projectors. Complex phases permit two distinct nonorthogonal two-ray decompositions of the same positive operator; over the reals, its two simple eigenvalues fix the rays. The 21-vertex, 14-context hypergraph H_{21} completes both inner pairs with a common ray. The projector spectrum then becomes degenerate and no longer yields a contradiction, but compatibility with the octagonal perimeter supplies a second, cyclic-closure obstruction.

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The full transition table gives these incidence results an operational role. For each configuration we compare its ideal complex table with all reciprocal real-qutrit prepare-and-test models. The real model space is compact. Since an exact fit would be a faithful real orthogonal representation, which our no-go theorems exclude, every real model has a strictly positive uniform error. The particularly simple H_{21} table has off-diagonal entries only in $\{0, 1/4, 1/2\}$. It therefore defines a finite, calibrated transition-probability witness of complex qutrit geometry. We state this operational consequence precisely in Sec. V; the optimal numerical robustness and the smallest sufficient subset of transitions remain open optimization problems.

Neither witness relies on KS uncolorability. The separation instead uses orthogonality, completeness of qutrit contexts, and target overlaps strictly between zero and one for all nonedges. We verify every unordered pair exactly: prescribed pairs are orthogonal, every other inner product is nonzero, and distinct vertices label distinct rays. Thus "faithful" has its full graph-theoretic meaning [12, 13] and, operationally, every edge and nonedge is represented in the transition table.

Finally, phase-adjusted realification [14] faithfully represents the underlying orthogonality data in $\mathbb{R}^6$. This is an upper bound, not a proven minimum: $\mathbb{R}^4$ and $\mathbb{R}^5$ are not addressed. It generally changes nonzero transition probabilities, and each three-ray context spans only half the real space. We give explicit realifications below.

II. REPRESENTATIONS AND ALGEBRAIC TOOLS

Throughout, we use the physics convention for the Hermitian inner product, which is conjugate-linear in the first argument:

$$\langle x, y\rangle = \sum_{k=1}^{d} \overline{x_k}\, y_k. \tag{2}$$

Definition 1. Let $\mathsf{H} = (V, \mathcal{E})$ be a three-uniform hypergraph, let $\mathbb{F}$ be either $\mathbb{R}$ or $\mathbb{C}$, and let $d \geq 3$. A *faithful orthogonal representation* of H in $\mathbb{F}^d$ assigns a nonzero vector $x_v \in \mathbb{F}^d$ to every $v \in V$ such that:

1. distinct vertices determine distinct rays; and
2. $\langle x_u, x_v\rangle = 0$ if and only if u and v belong to a common hyperedge.

Each hyperedge is then an orthogonal triple. For $d = 3$ it is an orthogonal basis and hence a maximal *context*; for $d > 3$ the same specified triple need not be maximal.

For a nonzero column vector x, let $x^\dagger$ denote its conjugate transpose and write

$$P_x = \frac{x x^\dagger}{\langle x, x\rangle} \tag{3}$$

for the rank-one projector onto its ray. In a three-dimensional representation, a context $\{x, y, z\}$ satisfies

$$P_x + P_y + P_z = I_3. \tag{4}$$

Consequently, integer linear combinations of three-dimensional context equations can force projector identities that hold in every representation of the hypergraph in that dimension.

The transition probability between two reciprocal pure preparations and rank-one tests is their squared normalized overlap

$$Q(x, y) := \mathrm{Tr}(P_x P_y) = \frac{|\langle x, y\rangle|^2}{\langle x, x\rangle\langle y, y\rangle}. \tag{5}$$

Thus $Q = 0$ exactly for orthogonal rays and $Q = 1$ exactly for coincident rays. Unlike the zero pattern alone, the complete table of Q-values also records every nonedge of a faithful orthogonality representation.

For $x, y \in \mathbb{C}^3$ define the Hermitian cross product by

$$x \boxtimes y := \overline{x \times y}, \tag{6}$$

where $x \times y$ is the ordinary bilinear cross product. Then $x \boxtimes y$ is Hermitian-orthogonal to both x and y.

Lemma 2 (Cyclic Gram closure)**.** *Let $x, y, z \in \mathbb{C}^3$, and suppose the two cross products in Eq. (6) are nonzero. Then*

$$\langle x \boxtimes y, y \boxtimes z\rangle = 0 \iff \langle x, y\rangle\langle y, z\rangle = \langle x, z\rangle\langle y, y\rangle. \tag{7}$$

Proof. The bilinear Lagrange identity, with $\cdot$ denoting the standard bilinear dot product on $\mathbb{C}^3$, gives

$$(x \times y) \cdot (\bar{y} \times \bar{z}) = (x \cdot \bar{y})(y \cdot \bar{z}) - (x \cdot \bar{z})(y \cdot \bar{y}).$$

Indeed, for arbitrary $a, b \in \mathbb{C}^3$, $\langle \bar{a}, \bar{b}\rangle = a \cdot \bar{b}$; taking $a = x \times y$ and $b = y \times z$ therefore identifies the left-hand side with $\langle x \boxtimes y, y \boxtimes z\rangle$. The right-hand side is the complex conjugate of

$$\langle x, y\rangle\langle y, z\rangle - \langle x, z\rangle\langle y, y\rangle.$$

Their vanishing is therefore equivalent. □

Lemma 3 (Rigidity of two real rays)**.** *Let a_1, a_2, b_1, b_2 be real unit vectors satisfying*

$$P_{a_1} + P_{a_2} = P_{b_1} + P_{b_2}. \tag{8}$$

If a_1 and a_2 are distinct and nonorthogonal, then $\{[b_1], [b_2]\} = \{[a_1], [a_2]\}$ as unordered pairs of rays.

Proof. All four vectors lie in the range of the common operator in Eq. (8). Choose signs so that $\gamma = \langle a_1, a_2\rangle \in (0, 1)$. The two nonzero eigenvalues are $1 + \gamma$ and $1 - \gamma$, with eigenvectors $a_1 + a_2$ and $a_1 - a_2$, respectively. Choose signs and order for the b-vectors so that $\delta = \langle b_1, b_2\rangle \geq 0$.

Equality of spectra gives $\delta = \gamma$. Since the two eigenvalues are distinct, there are real nonzero scalars λ, μ such that

$$b_1 + b_2 = \lambda(a_1 + a_2), \qquad b_1 - b_2 = \mu(a_1 - a_2).$$

The unit-norm and inner-product equations give

$$\begin{aligned}\lambda^2(1+\gamma) + \mu^2(1-\gamma) &= 2,\\ \lambda^2(1+\gamma) - \mu^2(1-\gamma) &= 2\gamma.\end{aligned}$$

Hence $\lambda^2 = \mu^2 = 1$. Solving for b_1, b_2 recovers the two original rays, possibly with signs and permutation. □

III. THE 20-RAY OCTAGON

The vertices of H_{20} are eight midpoint vertices $v_0, \ldots, v_7$, eight corner vertices $v_{01}, v_{12}, \ldots, v_{70}$, and four inner vertices a_1, a_2, b_1, b_2. Indices are read modulo eight. The eight perimeter contexts and four inner contexts are

$$E_j = \{v_{j-1,j}, v_j, v_{j,j+1}\}, \qquad j = 0, \ldots, 7, \tag{9}$$

$$\begin{aligned} D_0 &= \{v_0, a_1, v_4\}, & D_2 &= \{v_2, a_2, v_6\},\\ D_1 &= \{v_1, b_2, v_5\}, & D_3 &= \{v_3, b_1, v_7\}. \end{aligned} \tag{10}$$

The incidence structure is shown in Fig. 1.

A. The covering identity

Consider the two collections of six contexts

$$\begin{aligned}\mathcal{A} &= \{D_0, D_2, E_1, E_3, E_5, E_7\},\\ \mathcal{B} &= \{D_1, D_3, E_0, E_2, E_4, E_6\}.\end{aligned}$$

Every midpoint and every corner occurs exactly once in each collection. Only the two a-vertices occur in $\mathcal{A}$, and only the two b-vertices occur in $\mathcal{B}$. Summing Eq. (4) over the two collections and subtracting therefore gives

$$P_{a_1} + P_{a_2} = P_{b_1} + P_{b_2}. \tag{11}$$

This identity is combinatorial: it holds in every real or complex three-dimensional representation of H_{20}.

B. An exact faithful representation over $\mathbb{C}^3$

Set

$$\begin{aligned} C &= \frac{49 - 5\sqrt{69}}{26}, & s &= \sqrt{C},\\ N &= 1 + C, & r &= \sqrt{N}. \end{aligned} \tag{12}$$

The number C is the smaller positive root of

$$13C^2 - 49C + 13 = 0, \tag{13}$$

so $0 < C < 1$. Define the unit phases

$$\eta = \frac{5 + 12\mathrm{i}}{13}, \tag{14}$$

$$\zeta = \frac{14 + 6\sqrt{69}}{65} + \mathrm{i}\frac{4\sqrt{69} - 21}{65}. \tag{15}$$

Let

$$\begin{aligned}(\mu_0, \ldots, \mu_7) &= (-1, \eta, 1, -\eta, -1, \eta, 1, -\eta),\\ (\tau_0, \ldots, \tau_7) &= (1, \zeta, -\mathrm{i}, -\mathrm{i}\zeta, -1, -\zeta, \mathrm{i}, \mathrm{i}\zeta).\end{aligned}$$

The inner and midpoint rays are represented by

$$\begin{aligned} a_1 &= (s, 1, 0), & a_2 &= (s, -1, 0),\\ b_1 &= (s, \eta, 0), & b_2 &= (s, -\eta, 0), \end{aligned} \tag{16}$$

$$v_j = (1, s\mu_j, r\tau_j), \qquad j = 0, \ldots, 7. \tag{17}$$

For j modulo eight, define

$$v_{j,j+1} = v_j \boxtimes v_{j+1}. \tag{18}$$

Equivalently, Eq. (18) is the explicit formula

$$\begin{aligned} v_{j,j+1} = \big(&sr(\bar{\mu}_j\bar{\tau}_{j+1} - \bar{\tau}_j\bar{\mu}_{j+1}),\\ &r(\bar{\tau}_j - \bar{\tau}_{j+1}), s(\bar{\mu}_{j+1} - \bar{\mu}_j)\big). \end{aligned} \tag{19}$$

Theorem 4. *Equations (12)–(19) give a faithful orthogonal representation of H_{20} in $\mathbb{C}^3$.*

Proof. First, $|\eta| = 1$. Write $\zeta = x + \mathrm{i}y$. The four inner contexts follow directly from $r^2 = N$, $|\eta| = |\zeta| = 1$, and the antipodal relations $\mu_{j+4} = \mu_j$, $\tau_{j+4} = -\tau_j$. For example,

$$\langle a_1, v_0\rangle = 0, \qquad \langle v_0, v_4\rangle = 1 + C - r^2 = 0,$$

and

$$\langle b_2, v_1\rangle = 0, \qquad \langle v_1, v_5\rangle = 1 + C - r^2|\zeta|^2 = 0.$$

The other two inner contexts are identical calculations.

By construction, $v_{j,j+1}$ is orthogonal to v_j and v_{j+1}. It remains to prove orthogonality of the two corner rays in every perimeter context. Put

$$G_{jk} = \langle v_j, v_k\rangle = 1 + C\bar{\mu}_j\mu_k + N\bar{\tau}_j\tau_k.$$

Lemma 2 reduces the eight remaining conditions to

$$G_{j-1,j}G_{j,j+1} = G_{j-1,j+1}G_{j,j}, \qquad j = 0, \ldots, 7. \tag{20}$$

Substituting the cyclic patterns for μ_j and τ_j reduces all eight conditions to common real and imaginary equations once $x^2 + y^2 = 1$. Separating those parts gives

$$7Cx - 17Cy - 13C - 13x + 13y + 13 = 0, \tag{21}$$

and

$$\begin{aligned} &-7C^2x + 17C^2y + 13C^2 - 20Cx + 30Cy\\ &\qquad + 2C - 13x + 13y + 13 = 0. \end{aligned} \tag{22}$$

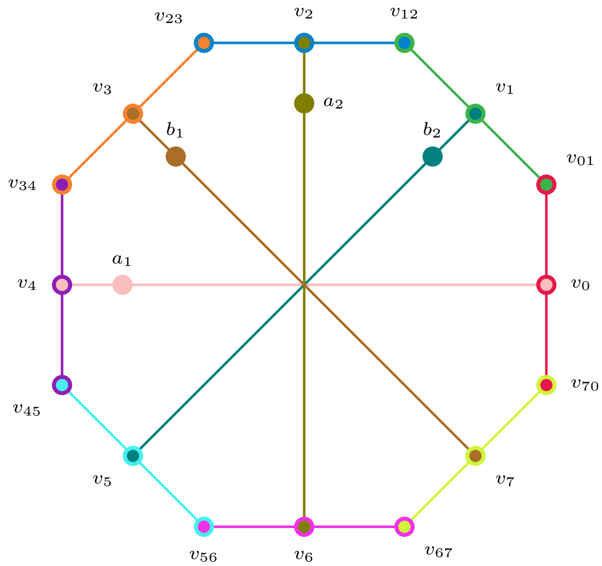

FIG. 1. The 20-ray hypergraph H_{20}. The eight perimeter contexts $E_0, \ldots, E_7$ form the octagon; the four inner contexts join opposite midpoints through a_1, a_2, b_1, b_2. Each color denotes one context.

Solving these two linear equations for x and y gives

$$x = \frac{104 - 76C}{65(1+C)}, \qquad y = \frac{39 - 81C}{65(1+C)}. \tag{23}$$

For these values,

$$x^2 + y^2 - 1 = \frac{48(13C^2 - 49C + 13)}{325(1+C)^2} = 0. \tag{24}$$

Using the smaller root in Eq. (13), Eq. (23) is exactly the phase in Eq. (15). Thus all twelve triples are contexts.

It remains to exclude unintended orthogonalities and ray coincidences. The exact overlap certificate in Appendix A shows that the 36 prescribed unordered pairs have zero overlap, whereas the 154 unprescribed pairs have squared normalized overlaps between 1/25 and 4/5. Hence every unprescribed inner product is nonzero and no two displayed vectors are proportional. The representation is faithful. □

The two inner pairs indeed decompose the same operator. Since all four vectors have squared norm N and $|\eta| = 1$,

$$P_{a_1} + P_{a_2} = P_{b_1} + P_{b_2} = \frac{2}{N}\,\mathrm{diag}(C, 1, 0). \tag{25}$$

Their within-pair inner products are $C - 1 \neq 0$, and the four rays are distinct because $\eta \notin \{1, -1\}$.

C. Nonexistence over $\mathbb{R}^3$

Theorem 5. *The hypergraph H_{20} has no faithful orthogonal representation in $\mathbb{R}^3$.*

Proof. Suppose a faithful real representation existed. The covering argument is independent of the field, so Eq. (11) would hold. Faithfulness implies that a_1, a_2 are distinct and nonorthogonal, because they do not share a context; the same is true of b_1, b_2. Lemma 3 then identifies the unordered pair of b-rays with the unordered pair of a-rays. This contradicts the requirement that four distinct vertices label four distinct rays. □

IV. THE 21-RAY ORTHOGONAL AUGMENTATION

Add one vertex c and two contexts

$$F_a = \{a_1, c, a_2\}, \qquad F_b = \{b_1, c, b_2\} \tag{26}$$

to H_{20}. The result is the 21-vertex, 14-context hypergraph H_{21} shown in Fig. 2. The added contexts force both inner pairs to be orthogonal and to have the same one-dimensional orthogonal complement.

A. Exact coordinates over $\mathbb{C}^3$

Let

$$p = \frac{\sqrt{14} + \sqrt{2}}{4}, \qquad q = \frac{\sqrt{14} - \sqrt{2}}{4}. \tag{27}$$

Then

$$p^2 + q^2 = 2, \qquad pq = \frac{3}{4}. \tag{28}$$

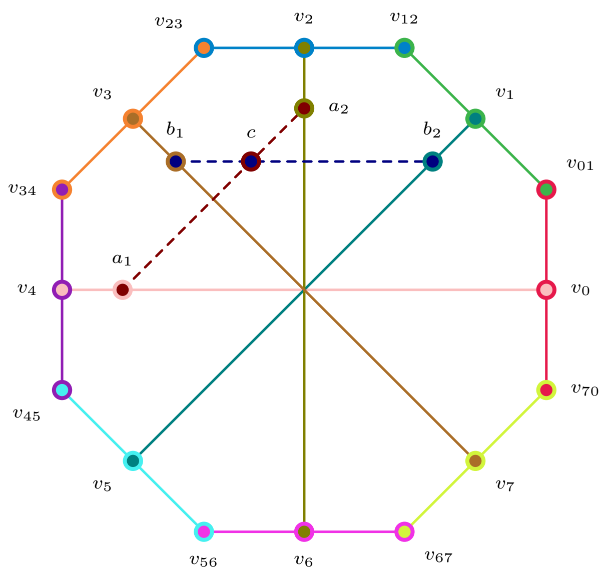

FIG. 2. The 21-ray hypergraph H_{21}. The twelve solid contexts are those of H_{20}; the two dashed contexts meet at the new vertex c.

Choose

$$\begin{aligned} c &= (0,0,1), \\ a_1 &= (1,1,0), & a_2 &= (1,-1,0), \\ b_1 &= (1,\mathrm{i},0), & b_2 &= (1,-\mathrm{i},0), \end{aligned} \tag{29}$$

and the eight midpoint representatives

$$\begin{aligned} v_0 &= (1,-1,\sqrt{2}), & v_4 &= (1,-1,-\sqrt{2}), \\ v_1 &= (1,\mathrm{i},p+\mathrm{i}q), & v_5 &= (1,\mathrm{i},-p-\mathrm{i}q), \\ v_2 &= (1,1,-\mathrm{i}\sqrt{2}), & v_6 &= (1,1,\mathrm{i}\sqrt{2}), \\ v_3 &= (1,-\mathrm{i},q-\mathrm{i}p), & v_7 &= (1,-\mathrm{i},-q+\mathrm{i}p). \end{aligned} \tag{30}$$

The eight corner representatives are

$$\begin{aligned} v_{01} &= (-p+\mathrm{i}(q+\sqrt{2}),\ \sqrt{2}-p+\mathrm{i}q,\ 1-\mathrm{i}), \\ v_{12} &= (\sqrt{2}-p+\mathrm{i}q,\ p-\mathrm{i}(q+\sqrt{2}),\ 1+\mathrm{i}), \\ v_{23} &= (q+\sqrt{2}+\mathrm{i}p,\ -q+\mathrm{i}(\sqrt{2}-p),\ -1+\mathrm{i}), \\ v_{34} &= (q+\mathrm{i}(p-\sqrt{2}),\ q+\sqrt{2}+\mathrm{i}p,\ -1-\mathrm{i}), \\ v_{45} &= (p-\mathrm{i}(q+\sqrt{2}),\ -\sqrt{2}+p-\mathrm{i}q,\ 1-\mathrm{i}), \\ v_{56} &= (-\sqrt{2}+p-\mathrm{i}q,\ -p+\mathrm{i}(q+\sqrt{2}),\ 1+\mathrm{i}), \\ v_{67} &= (-q-\sqrt{2}-\mathrm{i}p,\ q-\mathrm{i}(\sqrt{2}-p),\ -1+\mathrm{i}), \\ v_{70} &= (-q-\mathrm{i}(p-\sqrt{2}),\ -q-\sqrt{2}-\mathrm{i}p,\ -1-\mathrm{i}). \end{aligned} \tag{31}$$

Theorem 6. *Equations (27)–(31) give a faithful orthogonal representation of H_{21} in $\mathbb{C}^3$.*

Proof. The two added contexts are immediate from Eq. (29). For the original inner contexts, representative calculations are

$$\begin{aligned} \langle a_1, v_0\rangle &= 0, & \langle v_0, v_4\rangle &= 1+1-2 = 0, \\ \langle b_2, v_1\rangle &= 0, & \langle v_1, v_5\rangle &= 2-(p^2+q^2) = 0. \end{aligned}$$

The two remaining inner contexts follow in the same way.

Direct evaluation of the Hermitian cross product gives, exactly,

$$v_{j,j+1} = v_j \boxtimes v_{j+1}$$

for each representative in Eq. (31). Hence every corner is orthogonal to its two incident midpoints. Substitution in the Gram closure identity (7) reduces the eight remaining perimeter inner products to the two relations in Eq. (28); they therefore vanish.

Finally, Appendix A gives an exact check of all 210 unordered pairs. The 42 pairs prescribed by the fourteen contexts have zero overlap. Every remaining pair has squared normalized overlap either $1/4$ or $1/2$. Thus there are no additional orthogonalities and no ray coincidences. □

B. Nonexistence over $\mathbb{R}^3$

Theorem 7. *The hypergraph H_{21} has no faithful orthogonal representation in $\mathbb{R}^3$.*

Proof. Assume that a faithful real representation exists. A global orthogonal transformation and independent rescalings of ray representatives allow us to set

$$c = (0,0,1), \qquad a_1 = (1,0,0), \qquad a_2 = (0,1,0). \tag{32}$$

The second added context lies in the plane $c^\perp$, so for some angle θ,

$$b_1 = (\cos\theta, \sin\theta, 0), \qquad b_2 = (-\sin\theta, \cos\theta, 0). \tag{33}$$

Faithfulness excludes $\theta \in \frac{\pi}{2}\mathbb{Z}$, since those values identify a b-ray with an a-ray. Put

$$m = \tan\theta \neq 0, \qquad M = 1+m^2. \tag{34}$$

Write a representative vector of a midpoint ray as $v_j = (X_j, Y_j, Z_j)$. No midpoint ray can lie in $c^\perp$. Indeed, a planar vector orthogonal to the inner ray in its context is proportional to the other member of one of the two planar bases $\{a_1, a_2\}$ or $\{b_1, b_2\}$, contradicting faithfulness. Thus $Z_j \neq 0$ for every j, and each representative may be rescaled to

$$v_j = (X_j, Y_j, 1). \tag{35}$$

The inner contexts D_0, D_1, D_3 imply

$$X_0 = X_4 = 0, \quad Y_0 Y_4 + 1 = 0, \tag{36}$$
$$Y_1 = mX_1, \qquad Y_5 = mX_5, \qquad MX_1X_5 + 1 = 0, \tag{37}$$
$$X_3 = -mY_3, \qquad X_7 = -mY_7, \qquad MY_3Y_7 + 1 = 0. \tag{38}$$

In particular,

$$X_5 = -\frac{1}{MX_1}, \qquad Y_7 = -\frac{1}{MY_3}, \tag{39}$$

and $X_1Y_3 \neq 0$.

The corner rays can be eliminated from the perimeter by the real form of Lemma 2. At $j = 0$, Eq. (7) factors as

$$Y_0\big((mX_1Y_7 - 1)Y_0 + Y_7 + mX_1\big) = 0. \tag{40}$$

The first factor cannot vanish because of Eq. (36). Moreover, the coefficient $1 - mX_1Y_7$ cannot vanish: otherwise Eq. (40) would also give $Y_7 + mX_1 = 0$, and the two relations would imply $-m^2X_1^2 = 1$. Therefore

$$Y_0 = \frac{Y_7 + mX_1}{1 - mX_1Y_7}. \tag{41}$$

The closure equation at $j = 4$ gives analogously

$$Y_4 = \frac{Y_3 + mX_5}{1 - mY_3X_5}, \tag{42}$$

with a nonzero denominator.

Substituting Eq. (39) into Eqs. (41) and (42) yields

$$Y_0 = \frac{mMX_1Y_3 - 1}{MY_3 + mX_1}, \qquad Y_4 = \frac{MX_1Y_3 - m}{MX_1 + mY_3}. \tag{43}$$

Set $u = X_1Y_3$. The remaining equation $Y_0Y_4 = -1$ becomes

$$(mMu - 1)(Mu - m) = -(mX_1 + MY_3)(MX_1 + mY_3). \tag{44}$$

Expanding the left-hand side gives

$$\begin{aligned}(mMu - 1)(Mu - m) &= mM^2u^2 - M(m^2 + 1)u + m \\ &= mM^2u^2 - M^2u + m,\end{aligned}$$

where the second equality uses $M = 1 + m^2$. The right-hand side is

$$-u(M^2 + m^2) - mM(X_1^2 + Y_3^2).$$

After cancellation and division by $m \neq 0$, Eq. (44) is therefore equivalent to the master equation

$$M^2u^2 + mu + 1 + M(X_1^2 + Y_3^2) = 0. \tag{45}$$

Since X_1, Y_3 are real, $X_1^2 + Y_3^2 \geq 2|X_1Y_3| = 2|u|$. The left-hand side of Eq. (45) is consequently at least

$$\begin{aligned}M^2u^2 + mu + 1 + 2M|u| &\geq 1 + M^2u^2 + |u|(2M - |m|) \\ &\geq 1.\end{aligned}$$

Here, with $t = |m|$,

$$2M - |m| = 2t^2 - t + 2 = 2\left(t - \frac{1}{4}\right)^2 + \frac{15}{8} > 0.$$

This contradicts Eq. (45), and the assumed real representation cannot exist. $\square$

V. A FINITE TRANSITION-PROBABILITY FIELD WITNESS

We now turn the two exact representability separations into a calibrated prepare-and-test task. For $n \in \{20, 21\}$, let V_n be the vertex set of H_n, let $\psi_v^{(n)} \in \mathbb{C}^3$ be the complex representative displayed above, and define the ideal transition table

$$T_{uv}^{(n)} = Q\big(\psi_u^{(n)}, \psi_v^{(n)}\big), \qquad u, v \in V_n. \tag{46}$$

Operationally, label v specifies both the preparation $\rho_v = P_{\psi_v^{(n)}}$ and the reciprocal yes–no test $E_v = P_{\psi_v^{(n)}}$. The ideal accepting probability is then $p(v|u) = \mathrm{Tr}(\rho_u E_v) = T_{uv}^{(n)}$.

To compare with real qutrits, fix any ordering of V_n and define the uniform real fitting error

$$\varepsilon_{\mathbb{R}}^{(n)} = \min_{\substack{r_v \in \mathbb{R}^3,\ \|r_v\| = 1 \\ v \in V_n}} \max_{u < v} \left|(r_u^\mathsf{T} r_v)^2 - T_{uv}^{(n)}\right|. \tag{47}$$

This comparison is dimension bounded and calibrated: each test is assumed to be the rank-one projector associated with the correspondingly labeled pure preparation. It is not a device-independent Bell or contextuality scenario.

Theorem 8 (Positive real-qutrit gap)**.** *The ideal tables $T^{(20)}$ and $T^{(21)}$ are realized exactly by complex qutrits, whereas*

$$\varepsilon_{\mathbb{R}}^{(20)} > 0, \qquad \varepsilon_{\mathbb{R}}^{(21)} > 0. \tag{48}$$

For H_{21}, every off-diagonal target probability belongs to $\{0, 1/4, 1/2\}$.

Proof. The displayed complex representatives realize Eq. (46), so the corresponding complex fitting error is zero. For the real problem, the domain $(S^2)^{|V_n|}$ in

Eq. (47) is compact and the objective is continuous; hence the minimum is attained.

Suppose that this minimum were zero. Then some family of real unit vectors would reproduce every target entry exactly. The target value is zero for precisely the pairs prescribed by the contexts and is strictly positive for every other pair. Moreover, for distinct vertices the target is strictly less than one: Appendix A gives the bound $T^{(20)}_{uv} \le 4/5$ and gives $T^{(21)}_{uv} \in \{1/4, 1/2\}$ for every nonorthogonal pair. The real vectors would therefore have exactly the prescribed orthogonalities, no ray coincidences, and no unintended orthogonalities. They would constitute a faithful orthogonal representation of H_n in $\mathbb{R}^3$, contradicting Theorem 5 for $n = 20$ and Theorem 7 for $n = 21$. Thus both real minima are strictly positive. The final statement follows from the exact overlap inventory in Appendix A. □

Theorem 8 supplies a finite robustness statement: every reciprocal real-qutrit model differs from the ideal table in at least one tested transition by at least $\varepsilon^{(n)}_{\mathbb{R}}$. Consequently, a calibrated experiment whose systematic and statistical uncertainty places the complete table within a smaller uniform error rules out that model class. The proof establishes a nonzero gap but does not determine its optimal value. Computing a certified lower bound, optimizing the weights in an averaged score, and finding the smallest subset of transitions that retains a gap are natural next steps toward an economical implementation. The H_{21} table is especially attractive for this purpose because its 210 unordered off-diagonal transition pairs have only three ideal probability values.

VI. FAITHFUL REALIFICATION IN SIX DIMENSIONS

Theorems 5 and 7 keep the ambient dimension fixed. When dimension doubling is allowed, both hypergraphs have faithful real representations. We first recall the finite construction and then give explicit phase choices for the present coordinates [14].

Theorem 9 (Phase-adjusted realification). *Every finite faithful orthogonal representation by rays in $\mathbb{C}^3$ induces a faithful orthogonal representation of the same hypergraph by rays in $\mathbb{R}^6$.*

Proof. Let $\psi_1, \dots, \psi_N \in \mathbb{C}^3$ be nonzero representatives and put $c_{k\ell} = \langle \psi_k, \psi_\ell \rangle$. Define the canonical realification

$$\Phi_0(z_1, z_2, z_3) = (\Re z_1, \Re z_2, \Re z_3, \Im z_1, \Im z_2, \Im z_3). \tag{49}$$

For phases $\theta_k \in \mathbb{R}$, set $R_k = \Phi_0(e^{\mathrm{i}\theta_k}\psi_k)$. Direct expansion gives

$$R_k \cdot R_\ell = \Re\left(e^{\mathrm{i}(\theta_\ell - \theta_k)} c_{k\ell}\right). \tag{50}$$

Thus $c_{k\ell} = 0$ implies $R_k \cdot R_\ell = 0$ for every phase choice. If $c_{k\ell} \ne 0$, write $c_{k\ell} = |c_{k\ell}| e^{\mathrm{i}\varphi_{k\ell}}$. The right-hand side of Eq. (50) vanishes precisely when

$$\theta_\ell - \theta_k + \varphi_{k\ell} \equiv \frac{\pi}{2} \pmod{\pi}. \tag{51}$$

Choose $\theta_1, \dots, \theta_N$ inductively. Suppose that the first $m-1$ phases have been fixed. For each $k < m$ with $c_{km} \ne 0$, the unwanted orthogonality equation $R_k \cdot R_m = 0$ excludes, by Eq. (51), exactly two values of θ_m modulo 2π. Their union is finite, so a phase outside it exists. Continuing through $m = N$ makes the real dot product nonzero for every originally nonorthogonal pair, while all original orthogonalities remain zero.

It remains only to check ray distinctness. If R_k and R_ℓ were real proportional, then $e^{\mathrm{i}\theta_k}\psi_k$ and $e^{\mathrm{i}\theta_\ell}\psi_\ell$ would be real proportional and hence would represent the same complex ray. Faithfulness of the original representation excludes this. Therefore the realified representation is faithful. □

For explicit coordinates it is convenient to parametrize the phase by a real number t:

$$\lambda_t = \frac{1 + \mathrm{i}t}{\sqrt{1+t^2}}, \qquad \Phi_t(z) = \Phi_0(\lambda_t z). \tag{52}$$

Writing $z = x + \mathrm{i}y$ with $x, y \in \mathbb{R}^3$ gives the entirely real formula

$$\Phi_t(x + \mathrm{i}y) = \frac{(x - ty,\ y + tx)}{\sqrt{1+t^2}} \in \mathbb{R}^6. \tag{53}$$

Moreover,

$$\Phi_{t_u}(u) \cdot \Phi_{t_v}(v) = \Re\left(\frac{(1 - \mathrm{i}t_u)(1 + \mathrm{i}t_v)}{\sqrt{(1+t_u^2)(1+t_v^2)}} \langle u, v \rangle\right). \tag{54}$$

Corollary 10 (Explicit real six-dimensional coordinates). *Let $\psi^{(20)}_v$ denote the representatives in Eqs. (12)–(19). Define*

$$t^{(20)}_v = \begin{cases} 1, & v \in \{v_{34}, v_{45}, v_{56}\}, \\ 2, & v \in \{v_{67}, v_{70}\}, \\ 0, & \textit{otherwise.} \end{cases} \tag{55}$$

Then $R^{(20)}_v = \Phi_{t^{(20)}_v}(\psi^{(20)}_v)$ is a faithful orthogonal representation of H_{20} in $\mathbb{R}^6$.

Likewise, let $\psi^{(21)}_v$ denote the representatives in Eqs. (27)–(31), and define

$$t^{(21)}_v = \begin{cases} 1, & v = v_{45}, \\ 2, & v \in \{v_2, v_3, v_6, v_7, v_{34}, v_{56}, v_{70}\}, \\ 3, & v = v_{67}, \\ 0, & \textit{otherwise.} \end{cases} \tag{56}$$

Then $R^{(21)}_v = \Phi_{t^{(21)}_v}(\psi^{(21)}_v)$ is a faithful orthogonal representation of H_{21} in $\mathbb{R}^6$.

Proof. Equations (53), (55), and (56) give all six real coordinates directly from the complex vectors already displayed. Exact substitution in Eq. (54) gives zero for exactly 36 of the $\binom{20}{2} = 190$ pairs in H_{20}, namely the pairs prescribed by its twelve contexts. The other 154 real dot products are nonzero. For H_{21}, exactly 42 of the $\binom{21}{2} = 210$ real dot products vanish, namely the pairs prescribed by its fourteen contexts, and the other 168 are nonzero. These calculations take place in the algebraic fields generated by the radicals in the complex coordinates and by $\sqrt{1+t^2}$; no numerical tolerance is involved. Ray distinctness follows from Theorem 9. $\square$

Each displayed three-ray context is therefore an orthogonal triple in $\mathbb{R}^6$, but it spans only a three-dimensional subspace. Its projector sum is the rank-three projector onto that subspace, not I_6. This loss of maximality is precisely why the real six-dimensional representations do not contradict the two real three-dimensional no-go theorems.

VII. DISCUSSION

The two no-go proofs expose different field-dependent mechanisms. For H_{20}, the context covering forces a rank-two positive operator to have two disjoint, nonorthogonal decompositions into rank-one projectors. Complex relative phases make this possible. Over the reals, the two simple eigenvalues determine the two rays, producing the rigidity of Lemma 3. For H_{21}, the inner pairs are orthogonal and their common projector sum is the identity on $c^{\perp}$. The spectral argument is therefore unavailable. Compatibility with the octagonal perimeter instead produces the positive master expression in Eq. (45). The second witness is not merely the orthogonal limit of the first proof; it requires a genuinely different obstruction.

The transition-probability formulation makes explicit what must be observed in addition to the contexts. Orthogonality supplies the zero probabilities, while probabilities strictly between zero and one on every unprescribed pair exclude both unintended orthogonalities and ray coincidences. In this calibrated setting, faithfulness is therefore not an unmeasured graph-theoretic convention: it is encoded by the complete table. For H_{21} the ideal experiment is particularly discrete, because every off-diagonal probability is 0, 1/4, or 1/2.

The compactness argument in Theorem 8 shows that the field separation survives a finite uniform perturbation. This conclusion is qualitative until a certified value of $\varepsilon_{\mathbb{R}}^{(n)}$ is obtained. The present witness is also not device independent: it assumes pure three-dimensional preparations, reciprocal rank-one projective tests, and a calibration identifying equally labeled preparations and tests. These assumptions distinguish the result from Bell tests of real versus complex quantum theory and from KS noncontextuality inequalities. They also make its scope precise: the result is a finite field witness for qutrit geometry, not a falsification of all real-Hilbert-space formulations.

Corollary 10 faithfully realizes the same orthogonality pattern in $\mathbb{R}^6$, where the triples are not complete measurements. The context sum is then a rank-three subspace projector, so neither the covering identity nor the three-dimensional cross-product closure follows. This removes the underlying hypergraph obstruction, but the rank-one real rays need not reproduce the nonzero table entries, and $\mathbb{R}^4$ and $\mathbb{R}^5$ remain unaddressed.

Faithfulness and the full incidence structure are essential: allowing extra orthogonalities or ray coincidences permits degeneracy, while deleting contexts weakens the projector identity or closure system. Thus the complete labeled vertex–context structure matters.

The next operational questions are quantitative. Polynomial optimization or semidefinite relaxations could provide certified lower bounds on $\varepsilon_{\mathbb{R}}^{(20)}$ and $\varepsilon_{\mathbb{R}}^{(21)}$. Optimizing a weighted average score may yield better statistical performance than the uniform norm in Eq. (47). It is also natural to search for a smaller subset of transitions that retains a real-complex gap, and ultimately for a self-testing or device-independent task generated by the same incidence mechanisms.

VIII. CONCLUSION

The configurations H_{20} and H_{21} define two exact finite transition-probability tables attained by complex qutrits and by no reciprocal real-qutrit prepare-and-test model. Their no-go proofs arise from two different mechanisms: rigidity of a two-projector decomposition and cyclic closure at the orthogonal limit. Exact pairwise certificates establish the complete tables without numerical or coloring assumptions; for H_{21} the off-diagonal probabilities are only 0, 1/4, and 1/2. Compactness then promotes exact nonrepresentability to a strictly positive real-qutrit approximation gap. The $\mathbb{R}^6$ rank-one realifications remove only the underlying orthogonality obstruction; they neither reproduce the complete tables nor determine the minimum real dimension. A certified numerical gap and a reduced transition set remain steps toward an efficient implementation.

Appendix A: Exact faithfulness certificates

Recall the squared normalized overlap Q from Eq. (5). All calculations below are exact reductions in the algebraic fields generated by the displayed radicals and phases. The reductions and pair counts were independently cross-checked by exact symbolic arithmetic; no floating-point tolerance is used.

For H_{20}, the 12 contexts prescribe 36 distinct unordered orthogonal pairs. Substitution of Eqs. (12)–(19), followed by reduction with Eq. (13), gives $Q = 0$ for precisely those 36 pairs. For the other 154 pairs, the complete list of values and multiplicities is:

Q	mult.	Q	mult.	Q	mult.
$\frac{1}{25}$	2	$\frac{3}{10}$	8	$\frac{8}{15}$	8
$\frac{3}{50}$	4	$\frac{23}{75}$	2	$\frac{14}{25}$	8
$\frac{7}{75}$	8	$\frac{49}{150}$	8	$\frac{16}{25}$	2
$\frac{8}{75}$	8	$\frac{26}{75}$	8	$\frac{7}{10}$	8
$\frac{7}{50}$	8	$\frac{28}{75}$	8	$\frac{59}{75}$	2
$\frac{4}{25}$	8	$\frac{32}{75}$	4	$\frac{4}{5}$	16
$\frac{1}{5}$	16	$\frac{7}{15}$	8		
$\frac{6}{25}$	8	$\frac{13}{25}$	2		

(A1)

The multiplicities sum to 154. In particular,

$$\frac{1}{25} \leq Q(x, y) \leq \frac{4}{5} \tag{A2}$$

for every unprescribed pair. This proves simultaneously that no such pair is orthogonal and that no two vertices label the same ray.

For H_{21}, the 14 contexts prescribe 42 distinct unordered orthogonal pairs. Exact substitution of Eqs. (28)–(31) gives zero for precisely those pairs. The other 168 pairs split as

Q	$\frac{1}{4}$	$\frac{1}{2}$
multiplicity	84	84

. (A3)

Again every unprescribed overlap lies strictly between zero and one. This completes the exact faithfulness verification used in Theorems 4 and 6.

DATA AVAILABILITY

No external data were used. The hypergraphs, exact coordinates, and complete pairwise overlap inventories needed to reproduce the reported results are contained in this article.

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